

AS 5401: Homework 2

Due Date: 12 Oct 2026

Instructions

- **Sign and attach this page to the first page of your submission. Submissions without this page signed and attached will not be graded**
- No late submissions will be received after the due date.
- Total marks is 100 (+ 30 bonus)
- Attach your code and plots to your submission.
- **Steps Followed to Obtain the Results:** Clearly outline each step you took to solve the problem, from setting up the code and input data, through analysis, to the final conclusions.
- **Analysis of Results:** A plot alone is not sufficient to obtain a grade—you must analyze and interpret the results. Your analysis should address:
 - a. How did you obtain this result?
 - b. What does the result mean?
 - c. Why is this result important in the context of the problem?
 - d. What are the advantages and disadvantages of the methods used?
- Your executable code should also be submitted.
- Make sure your computer code is well-commented, and include all relevant equations in your report to support your findings.

Honesty Code (To Be Signed and Submitted)

I attest that the assignment solutions I provided are my own and that I did not copy any solution from anybody in class or elsewhere. I further understand that if I am caught cheating on any homework or project solution, I am subject to the ethical violations procedures as outlined in the guidelines circulated for the quizzes: <https://drive.google.com/file/d/1FsL6pEcQvDBfmM-Euj4BP8-DhJuvRg8i/view>

Signature: _____

Name: _____

Roll No.: _____

Questions

1. **Controllability and Observability Gramians** Consider the continuous-time LTI system:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t), \quad \mathbf{y}(t) = \mathbf{C}\mathbf{x}(t)$$

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & -5 & -6 & -4 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{C} = [1 \quad 0 \quad 0 \quad 0].$$

- Compute The controllability Gramian $\mathbf{W}_c$ and observability Gramian $\mathbf{W}_o$ by solving the Lyapunov equations explicitly using the MATLAB command `>>lyap`
- Verify controllability and observability by computing the ranks of the controllability and observability matrices.
- Compute the same Gramians using the MATLAB command 'gram' (i.e., 'gram(sys,'c')', 'gram(sys,'o')') and confirm that both methods yield the same results.

2. Numerical Conditioning and Gramians

- Plot the eigenvalues of $\mathbf{W}_c$ and $\mathbf{W}_o$. Comment on the spread.
- Why does a large spread in eigenvalues indicate difficulty for balanced truncation?
- Perform a similarity transform that rescales the states and check how $\mathbf{W}_c$, $\mathbf{W}_o$ change. Report observations.

3. **Balanced Realization** If you follow these steps listed below, you will be implementing the balanced realization algorithm manually. Note that this is what is done manually to obtain the balanced realization. When you use *balreal* command in MATLAB it uses a different and more optimized algorithm. To investigate further, follow the steps listed below:

- Compute the balanced realization manually:
 - Compute Cholesky factors of $\mathbf{W}_c = \mathbf{R}_c \mathbf{R}_c^T$, $\mathbf{W}_o = \mathbf{R}_o \mathbf{R}_o^T$. Use the command
`R_c = chol(Wc, 'lower');`
`R_o = chol(Wo, 'lower');`
 - Perform SVD of $\mathbf{R}_c^T \mathbf{R}_o = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T$.
 - Construct the balancing transformation matrix $T = \mathbf{R}_c \mathbf{V} \mathbf{\Sigma}^{-\frac{1}{2}}$, $T^{-1} = \mathbf{\Sigma}^{-\frac{1}{2}} \mathbf{U}^T \mathbf{R}_o^T$
- Compare the transformation matrices obtained with MATLAB's 'balreal' command.
- Plot and compare the Hankel singular values.

4. Balanced Truncation Model Reduction

- From your balanced realization, truncate to order 2. Plot step responses of full vs reduced models.
- Use MATLAB's 'balred' for order 2 and compare.
- Compute the error bound from Hankel singular values.

5. **Balanced Truncation for structural vibration model** You are given the state-space model of a lightly damped mechanical system (a chain of six masses connected by springs and dampers). The model has to be converted to first-order form:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t), \quad \mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t),$$

where $\mathbf{x}(t) \in \mathbb{R}^{12}$, $\mathbf{u}(t)$ is a scalar input force applied to the last mass, and $\mathbf{y}(t)$ is the displacement vector of all six masses.

(a) **Stability check**

- Compute the eigenvalues of $\mathbf{A}$. Are all modes stable?
- Comment on whether the system is suitable for computing controllability and observability Gramians. Note that balanced truncation can be computed for systems with stable matrices only (as of what we learnt in class)

(b) **Controllability and Observability**

- Using MATLAB commands 'gram', compute the controllability and observability Gramians.
- Verify controllability and observability by checking the ranks of 'ctrb(A,B)' and 'obsv(A,C)'.
- Compare and comment: do the ranks match your expectations from the physical setup?

(c) **Balanced Realization**

- Use 'balreal' to compute the balanced realization and extract the Hankel singular values.
- Plot the Hankel singular values. Which states are most significant?
- Decide how many states can be truncated while still capturing the dominant input-output behavior.

(d) **Model Reduction via Balanced Truncation**

- Use 'balred' to reduce the system to order $r = 4$.
- Compare the step responses of the full-order model and the reduced-order model.
- Compute the relative $\mathcal{H}_\infty$ error between the two systems using " $\gg \text{norm}(\text{sys1} - \text{sys2}, \text{inf}) / \text{norm}(\text{sys1}, \text{inf})$ ".

(e) **Manual Balanced Truncation** Derive your own balanced truncation routine:

- Solve the Lyapunov equations for controllability and observability Gramians.
- Compute the balancing transformation manually.
- Compare your reduced model with MATLAB's 'balred' results.

6. **Dynamic Mode Decomposition** for cylinder flow data. You will be given the snapshot matrices of the vorticity field in flow past a cylinder. The data will be available in your class moodle page as `cylinder_flow_snapshots.mat`. The dataset is available on Moodle as `cylinder_flow_snapshots.mat`, which contains:

- 'X, Y': snapshot matrices of size ' $(n_x \times n_y) \times T$ ', with each column representing the vorticity field at a time step.
- Grid dimensions: ' n_x ', ' n_y '.
- A plotting function 'plotCylinder.m' is provided to visualize the flow field.
- Use this data to perform dynamic mode decomposition and obtain a reduced-order $\hat{\mathbf{A}}$ matrix such that

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{A}} \hat{\mathbf{x}}_k$$

- Ask explicitly how they determine the truncation rank (via singular values). That connects DMD to SVD/POD, making it deeper.
- Reconstruct the flowfield using the reduced-order operator obtained from dynamic mode decomposition and compare it to the full-order model. You will use the following command to obtain the cylinder flow plot at time t from snapshot matrix $\mathbf{X}$. The *plotCylinder* function will also be provided to you on your moodle page.

```
for t = 1:final_tym
    plotCylinder(reshape(real(X(:,t)),nx,ny),nx,ny)
    drawnow
end
```

- Plot errors for various reduced-order models size vs the full-order flow field using the appropriate norm. Justify your choice of norm and also provide insights on what you think is the right reduced-order model size.

7. [Bonus 30 marks] **Proper Orthogonal Decomposition (POD)-based Reduced-Order Model (ROM)**
In this assignment, you will implement a Proper Orthogonal Decomposition (POD)-based Reduced-Order Model (ROM) for the 1D viscous Burgers' equation:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}, \quad x \in [0, 1], \quad t > 0,$$

with Dirichlet boundary conditions $u(0, t) = u(1, t) = 0$ and an initial condition $u(x, 0) = \sin(\pi x)$.

You are **not required to derive the ROM equations**. The development is explained below, and your task is to implement the steps in code and compare the reduced solution with the full-order solution.

- (a) **Simulate Full-Order Model (FOM) to get data** We discretize the domain into N grid points with spacing Δx . Define the state vector

$$\mathbf{u}(t) = [u(x_1, t), u(x_2, t), \dots, u(x_N, t)]^T.$$

- First derivative (central difference):

$$\left(\frac{\partial u}{\partial x} \right)_i \approx \frac{u_{i+1} - u_{i-1}}{2\Delta x}.$$

- Second derivative:

$$\left(\frac{\partial^2 u}{\partial x^2} \right)_i \approx \frac{u_{i+1} - 2u_i + u_{i-1}}{(\Delta x)^2}.$$

Using derivative matrices $D_1, D_2 \in \mathbb{R}^{N \times N}$, the semi-discrete ODE system becomes:

$$\frac{d\mathbf{u}}{dt} = -\mathbf{u} \odot (D_1 \mathbf{u}) + \nu D_2 \mathbf{u},$$

where $\odot$ denotes elementwise multiplication.

- (b) **POD Approximation** We approximate the solution using POD modes via the methods of snapshots (involves SVD of the snapshot matrix). The approximation is given as:

$$\mathbf{u}(t) \approx \Phi \hat{\mathbf{u}}(t),$$

where

- $\Phi \in \mathbb{R}^{N \times r}$ contains the first r POD modes (orthonormal),
 - $\hat{\mathbf{u}}(t) \in \mathbb{R}^r$ are reduced coordinates.
- (c) * Intrusive projection based Reduced-Order Model (ROM)* Substituting the approximation into the FOM, we get:

$$\frac{d\hat{\mathbf{u}}}{dt} = \Phi^T \left(-(\Phi \hat{\mathbf{u}}) \odot (D_1(\Phi \hat{\mathbf{u}})) + \nu D_2(\Phi \hat{\mathbf{u}}) \right).$$

- Diffusion term (linear):

$$A = \nu \Phi^T D_2 \Phi, \quad A \in \mathbb{R}^{r \times r}.$$

- Convection term (quadratic, written in Kronecker form):

$$(\Phi \hat{\mathbf{u}}) \odot (D_1 \Phi \hat{\mathbf{u}}) = \text{diag}(\Phi \hat{\mathbf{u}})(D_1 \Phi \hat{\mathbf{u}}).$$

Expanding bilinearly in $\hat{\mathbf{u}}$:

$$(\Phi \hat{\mathbf{u}}) \odot (D_1 \Phi \hat{\mathbf{u}}) = \left[\text{diag}(\Phi)(D_1 \Phi) \right] (\hat{\mathbf{u}} \otimes \hat{\mathbf{u}}),$$

where $\otimes$ is the Kronecker product.

After projection:

$$N = -\Phi^T \left(\text{diag}(\Phi)(D_1 \Phi) \right), \quad N \in \mathbb{R}^{r \times r^2}.$$

Thus the convection contribution is:

$$N(\hat{\mathbf{u}} \otimes \hat{\mathbf{u}}).$$

(d) **Final ROM** The reduced system is:

$$\frac{d\hat{\mathbf{u}}}{dt} = A\hat{\mathbf{u}} + N(\hat{\mathbf{u}} \otimes \hat{\mathbf{u}}),$$

with

- $A \in \mathbb{R}^{r \times r}$: reduced diffusion operator,
- $N \in \mathbb{R}^{r \times r^2}$: reduced convection operator (matrix form),
- $\hat{\mathbf{u}} \otimes \hat{\mathbf{u}} \in \mathbb{R}^{r^2}$: Kronecker product of reduced state with itself.

(e) **Your Tasks**

- i. **FOM:** Discretize and solve the Burgers' equation for a chosen $\nu = 0.1$ and initial condition $u_0(x)$. Collect solution snapshots.
- ii. **POD:** Compute POD basis from the snapshot matrix using SVD. Select appropriate value for ROM size. What should be the size of ROM r . Justify your choice.
- iii. **ROM Assembly:**
 - Construct $A = \nu \Phi^T D_2 \Phi$.
 - Construct $N = -\Phi^T [\text{diag}(\Phi)(D_1 \Phi)]$.
- iv. **ROM Simulation:** Solve

$$\frac{d\hat{\mathbf{u}}}{dt} = A\hat{\mathbf{u}} + N(\hat{\mathbf{u}} \otimes \hat{\mathbf{u}})$$

for $\hat{\mathbf{u}}(t)$. Reconstruct the full state.

- v. **Validation:** Compare ROM vs FOM for different values of r . Plot errors and discuss.

(f) **Deliverables**

- i. MATLAB/Python code implementing FOM, POD, and ROM.
- ii. How will you select the size of the ROM, provide an explanation.
- iii. Plots of solutions (ROM vs FOM).
- iv. Error analysis for different r .
- v. Discussion of accuracy vs efficiency.

Note: The formulas for A and N are provided. You do not need to derive them — just implement and test.

Acknowledgement

The original cylinder flow data and required helper files have been obtained from the website <http://www.dmdbook.com>. It should be noted that the codes have been modified as per class requirements. The instructor thanks the authors for making the data public.